

RESEARCH & ANALYSIS

PORTFOLIO

PRESENTED BY HAZEM HAWARI



ABOUT ME

A Bachelor's graduate in Mathematics and Data Science from Universitas Andalas, specializing in Analysis and Geometry, equipped with a strong foundation in quantitative, conceptual, and analytical thinking. Technical competencies developed include data analysis and interpretation, mathematical modeling, algorithm development, as well as programming and data visualization. Problem-solving, leadership, cross-functional teamwork, and interpersonal communication skills have been honed through active involvement in various organizations and committees throughout university. Possesses a strong interest in risk analysis, financial management, and research, and is ready to be placed anywhere across Indonesia to contribute optimally. Committed to combining a data-driven approach with proven leadership skills to support strategic planning, risk management, and operational development within the company.

View My CV 



Andalas University – (Aug 2022 – Mar 2026)

*Bachelor of Science
Mathematics and Data Science
GPA: 3.48 / 4.00*

COURSERA – (May 2026 – Present)

*Sertifikat Profesional Analisis
Data IBM dengan Excel dan R*

RevoU – (May 2026 – Present)

*RevoU Mini Course Data
Analytics*

SMAN 2 Lubuk Basung – (2019 – 2022)

*Natural Science
GPA: 96.00 / 100*



Experiences

WORK

Mathematics and Data Science Computation Laboratory



1. Head of Upgrading and Recruitment Division (Dec 2024 - Dec 2025)
2. Laboratory Practicum Assistant (Jul 2024- Mar 2026)

Teaching Assistant



Served as a teaching assistant for the courses of Mathematical Statistics, Complex Functions, Linear Programming, and Mathematical English.

Organization

Pekan Seni Bermatematika (National Level)



1. Executive Chair PSB XXII (2025 - 2026)
2. Logistics Coordinator PSB XXI (2024)

BEM KM FMIPA, Andalas University



Advocacy and Student Welfare (2023)

Mathematics Student Association, Andalas University



1. Steering Committee of Musyawarah Anggota XXVIII (Feb 2026 - Present)
2. Logistics Coordinator of Musyawarah Anggota XXVII (2024)
3. Disciplinary Commission of Training Himatika Character (2024)
4. Logistics Coordinator of Himatika Fair (2023)

Research Title: Conducted quantitative financial analysis on technology sector stocks through index construction, statistical evaluation, and market trend analysis.

Brief Description:

Developed a custom stock market index by integrating and analyzing stock price data from major technology companies including META, AMZN, AAPL, NFLX, and GOOG. The project involved data cleaning, return calculation, descriptive statistical analysis, pivot table analysis, and index construction to evaluate market performance and stock behavior.

Topics & Expertise:

Technical Skills:

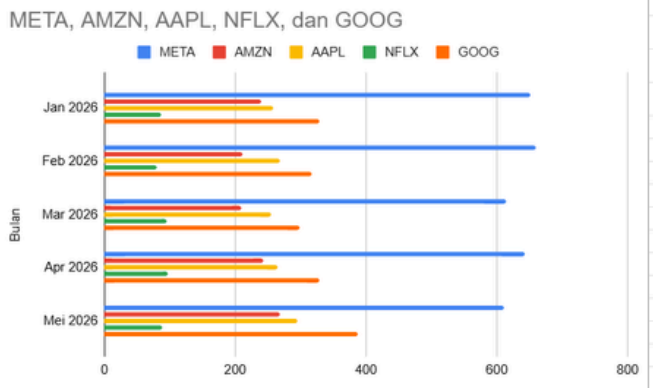
- Financial Data Analysis
- Stock Market Index Construction
- Data Cleaning & Validation
- Descriptive Statistics
- Quantitative Analysis
- Spreadsheet Modeling
- Return Calculation
- Pivot Table Analysis
- Statistical Interpretation
- Data Visualization

Tools:

- Google Sheet
- Pivot Tables
- Statistical Analysis
- Financial Modeling
- Data Cleaning Techniques



Bulan	META	AMZN	AAPL	NFLX	#	GOOG
Jan 2026	651,98	239,0845	257,6495	87,65	329,2805	
Feb 2026	659,1242105	212,0542105	269,1784211	80,53736842	316,9768421	
Mar 2026	614,9909091	209,995	254,9063636	95,07636364	297,9804545	
Apr 2026	642,1947619	242,8280952	264,6947619	97,80666667	328,7214286	
Mei 2026	611,0394444	268,3405556	295,9033333	88,31166667	388,3427778	



Column 1	Shares, bln	% Free floated shares, %	Free float,bln USD	% Weight
META	2,196046	18,85%	262,9688443	37,59%
AMZN	10,75711	24,36%	112,5335773	16,08%
AAPL	14,68736	33,84%	128,6778095	18,39%
NFLX	4,210799	17,13%	36,15894051	5,17%
GOOG	5,456	14,54%	159,302176	22,77%

Sheet	Total Rows	Valid Rows	Empty Rows	Error Count	Duplikat	Tanggal Mulai	Tanggal Akhir	Harga Negatif	Keterangan
META	100	100	0	0	0	1/2/2026	5/27/2026	0	✅ Data lengkap, tidak ada masalah
AMZN	100	100	0	0	0	1/2/2026	5/27/2026	0	✅ Data lengkap, tidak ada masalah
AAPL	100	100	0	0	0	1/2/2026	5/27/2026	0	✅ Data lengkap, tidak ada masalah
NFLX	100	100	0	0	0	1/2/2026	5/27/2026	0	✅ Data lengkap, tidak ada masalah
GOOG	100	100	0	0	0	1/2/2026	5/27/2026	0	✅ Data lengkap, tidak ada masalah

Date	META	AMZN	AAPL	NFLX	GOOG	META Return	AMZN Return	AAPL Return	NFLX Return	GOOG Return
1/2/2026	650,41	226,5	271,01	90,99	315,32	-	-	-	-	-
1/5/2026	658,79	233,06	267,26	91,46	317,32	1,29%	2,90%	-1,38%	0,52%	0,63%
1/6/2026	660,62	240,93	262,36	90,65	314,55	0,28%	3,38%	-1,83%	-0,89%	-0,87%
1/7/2026	648,69	241,56	260,33	90,73	322,43	-1,81%	0,26%	-0,77%	0,09%	2,51%
1/8/2026	646,06	246,29	259,04	90,53	326,01	-0,41%	1,96%	-0,50%	-0,22%	1,11%
1/9/2026	653,06	247,38	259,37	89,46	329,14	1,08%	0,44%	0,13%	-1,18%	0,96%
1/12/2026	641,97	246,47	260,25	89,41	332,73	-1,70%	-0,37%	0,34%	-0,06%	1,09%
1/13/2026	631,09	242,6	261,05	90,32	336,43	-1,69%	-1,57%	0,31%	1,02%	1,11%
1/14/2026	615,52	236,65	259,96	88,55	336,31	-2,47%	-2,45%	-0,42%	-1,96%	-0,04%
1/15/2026	620,8	238,18	258,21	88,05	333,16	0,86%	0,65%	-0,67%	-0,56%	-0,94%
1/16/2026	620,25	239,12	255,53	88	330,34	-0,09%	0,39%	-1,04%	-0,06%	-0,85%
1/20/2026	604,12	231	246,7	87,26	322,16	-2,60%	-3,40%	-3,46%	-0,84%	-2,48%
1/21/2026	612,96	231,31	247,65	85,36	328,38	1,46%	0,13%	0,39%	-2,18%	1,93%
1/22/2026	647,63	234,34	248,35	83,54	330,84	5,66%	1,31%	0,28%	-2,13%	0,75%
1/23/2026	658,76	239,16	248,04	86,12	328,43	1,72%	2,06%	-0,12%	3,09%	-0,73%
1/28/2026	672,36	238,42	255,41	85,7	333,59	2,06%	-0,31%	2,97%	-0,49%	1,57%
1/27/2026	672,97	244,68	258,27	85,58	335	0,09%	2,63%	1,12%	-0,14%	0,42%
1/28/2026	668,73	243,01	256,44	84,64	336,28	-0,63%	-0,68%	-0,71%	-1,10%	0,38%
1/29/2026	738,31	241,73	258,28	83,16	338,66	10,40%	-0,53%	0,72%	-1,75%	0,71%
1/30/2026	716,5	239,3	259,48	83,49	338,53	-2,95%	-1,01%	0,46%	0,40%	-0,04%
2/2/2026	706,41	242,96	270,01	82,76	344,9	-1,41%	1,53%	4,06%	-0,87%	1,88%
2/3/2026	691,7	238,62	269,48	79,94	340,7	-2,08%	-1,79%	-0,20%	-3,41%	-1,22%
2/4/2026	668,99	232,99	276,49	80,16	333,34	-3,28%	-2,36%	2,60%	0,28%	-2,16%

Statistik	META	AMZN	AAPL	NFLX	GOOG
Std Dev	36,58919072	23,8822699	15,8189158	7,099433797	32,28890677
Min	525,72	198,79	246,63	75,86	273,14
Max	738,31	274,99	310,85	107,79	399,04
Range	212,59	76,2	64,22	31,93	125,9
Q1	611,735	210,0825	257,205	85,67	306,23
Q3	660,83	251,69	273,4925	94,7325	339,725
IQR	49,095	41,6075	16,2875	9,0625	33,495
Daily Return					
Mean Return	330,72	0,00	0,00	0,00	0,00
Std Dev Return	32,42	0,03	0,02	0,02	0,02
Min Return	273,14	-0,09	-0,06	-0,05	-0,10
Max Return	399,04	0,10	0,06	0,04	0,14
Positif Days	99,00	53,00	55,00	53,00	46,00
Negatif Days	0,00	46,00	44,00	46,00	53,00
Win Rate	100,00%	53,54%	55,56%	53,54%	46,46%
BEST & WORST DAY					
Best Return	399,04	10,40%	5,60%	4,06%	13,77%
Best Date	5/13/2026	738,31	233,65	270,01	96,24
Worst Return	273,14	-8,55%	-5,55%	-5,00%	-9,72%
Worst Date	3/30/2026	611,91	210,32	261,73	97,31

Key Achievements:

- Constructed a custom stock market index using weighted stock data from major technology companies.
- Performed data cleaning and quality assessment to ensure complete, accurate, and consistent financial datasets.
- Calculated daily stock returns and analyzed market movement patterns across multiple assets.
- Applied descriptive statistical analysis including mean, median, standard deviation, quartiles, and range evaluation.
- Developed pivot table summaries to analyze monthly stock performance trends and price behavior.
- Demonstrated strong financial data analysis, quantitative reasoning, and spreadsheet modeling skills.

Click here for detail



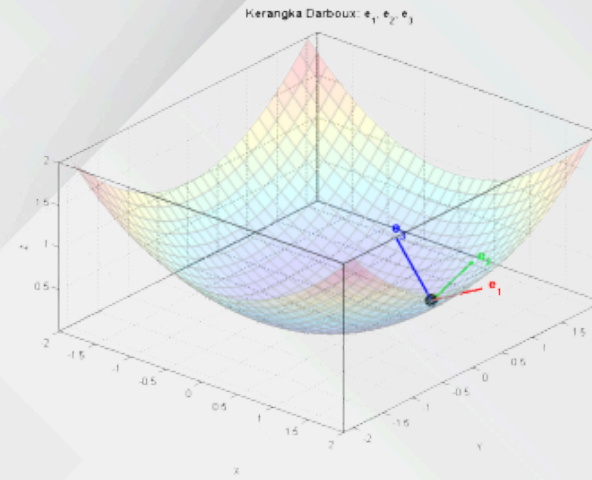
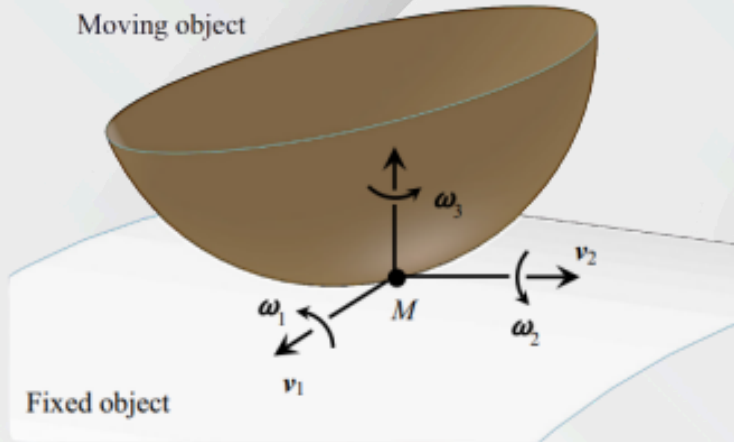
Research Title: Analysis of Translational and Rotational Velocity Models of Objects in Sliding-Rolling Motion Using an Adjoint Approach

Brief Description:

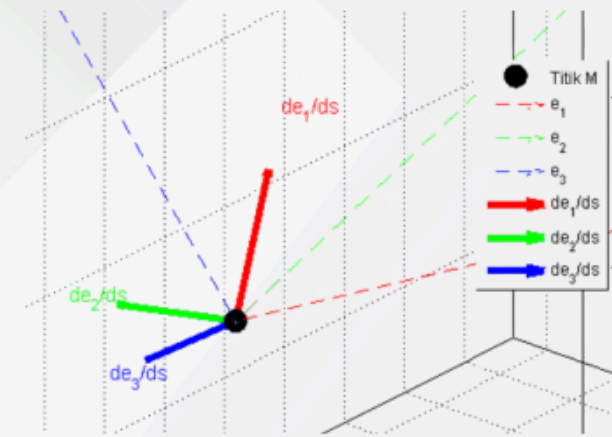
This research formulates a kinematic model of sliding-rolling motion between two regular surfaces using the adjoint approach and the Darboux framework of differential geometry. The main results are translational and rotational velocity equations that are invariant to the coordinate system, along with the contact point trajectory equations.

Topics & Expertise:

1. Differential geometry (curves, surfaces, Darboux frames, geodesic curvature)
2. Rigid body kinematics (roll-sliding, degrees of freedom)
3. Mathematical modeling and derivation of analytical equations
4. Numerical simulation using MATLAB
5. Visualization of kinematic data using Python



Gambar 2.7.1: Kerangka Darboux di sepanjang kurva pada permukaan



Gambar 2.7.2: Turunan vektor basis terhadap panjang busur s

$$\begin{cases} \frac{d\vec{r}_{OM}}{ds} = \vec{e}_1 \\ \frac{d}{ds} \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix} = \begin{bmatrix} 0 & \kappa_g & \kappa_n \\ -\kappa_g & 0 & \tau_g \\ -\kappa_n & -\tau_g & 0 \end{bmatrix} \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix} \end{cases}$$

Hasil Analisa Permukaan sehingga diperoleh model Kerangka Bergerak pada suatu titik

$$\vec{v} = (\sigma(1 - \lambda \cos \varphi))\vec{e}_1 - (\sigma \lambda \sin \varphi)\vec{e}_2,$$

$$\vec{\omega} = (\sigma(-\lambda(\kappa_n^* \sin \varphi + \tau_g^* \cos \varphi) + \tau_g))\vec{e}_1 + (\sigma(\lambda(\kappa_n^* \cos \varphi - \tau_g^* \sin \varphi) - \kappa_n))\vec{e}_2$$

$$+ (\sigma(-\lambda(\kappa_g^* - \frac{d\varphi}{ds^*}) + \kappa_g))\vec{e}_3.$$

(3.4.8)

Persamaan Kecepatan Translasi dan Rotasi dalam gerakan *Sliding-rolling*

$$x(s) = x_0 + \int_0^s [A_1(\gamma)(\vec{e}_1(\gamma) \cdot \vec{i}) + A_2(\gamma)(\vec{e}_2(\gamma) \cdot \vec{i}) + A_3(\gamma)(\vec{e}_3(\gamma) \cdot \vec{i})] d\gamma,$$

$$y(s) = y_0 + \int_0^s [A_1(\gamma)(\vec{e}_1(\gamma) \cdot \vec{j}) + A_2(\gamma)(\vec{e}_2(\gamma) \cdot \vec{j}) + A_3(\gamma)(\vec{e}_3(\gamma) \cdot \vec{j})] d\gamma, \quad (3.5.5)$$

$$z(s) = z_0 + \int_0^s [A_1(\gamma)(\vec{e}_1(\gamma) \cdot \vec{k}) + A_2(\gamma)(\vec{e}_2(\gamma) \cdot \vec{k}) + A_3(\gamma)(\vec{e}_3(\gamma) \cdot \vec{k})] d\gamma. \quad (3.5.6)$$

Persamaan Lintasan yang dihasilkan dalam gerakan Siliding-Rolling

Key Results Two velocity equations are derived:

1. Translational velocity: represents the sliding motion on the tangent plane of the contact point.
2. Rotational velocity: includes both rolling ($\vec{e}_1$ and $\vec{e}_2$ directions) and rotating ($\vec{e}_3$ /normal direction).
3. Equation of the trajectory of an arbitrary point on the moving surface in a fixed reference frame.
4. Validation using the classic example of a disc on a plane: it is shown that the velocity of the disc's center depends only on the sliding-rolling rate σ and is always parallel to the tangent path.

Click here for detail



Research Title: Analysis of Determining the Number and Position of Waypoints on Great Circle Routes Using an Elliptic Geometry Approach

Brief Description:

Developed a maritime route optimization model using elliptic geometry and genetic algorithms to determine optimal waypoint positions on Great Circle Routes (GCR). The study focused on minimizing travel distance and improving navigation efficiency between Teluk Bayur Port (Indonesia) and Port Elizabeth (South Africa).

Topics & Expertise:

Mathematical & Analytical Skills:

- Elliptic Geometry
- Differential Geometry
- Spherical Geometry
- Geodesic Analysis
- Mathematical Modeling
- Quantitative Analysis

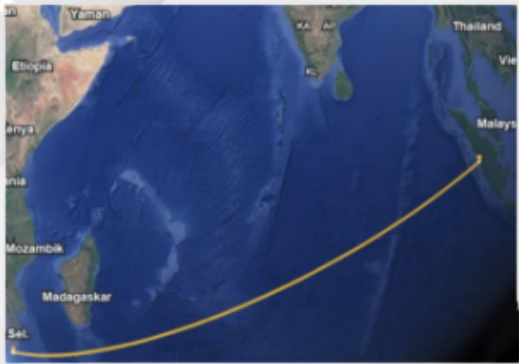
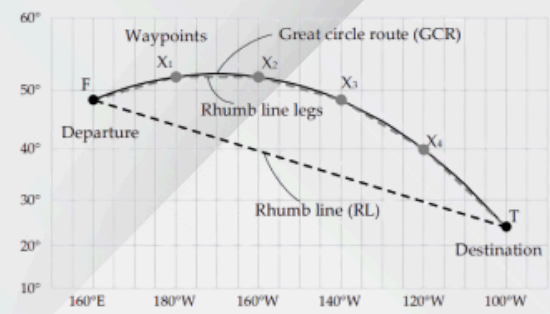
Computational & Optimization Skills:

- Genetic Algorithm (GA)
- Numerical Simulation
- Optimization Techniques
- Spatial Analysis
- Coordinate Transformation
- Scientific Computing

Methods & Techniques Used:

- Great Circle Route (GCR) Modeling
- Rhumb Line Approximation
- Genetic Algorithm Optimization
- Remaining Benefit (RB) Evaluation
- Numerical Simulation and Route Analysis

Gambar 1.1.1: Skema rute lingkaran besar dan *rhumb line*

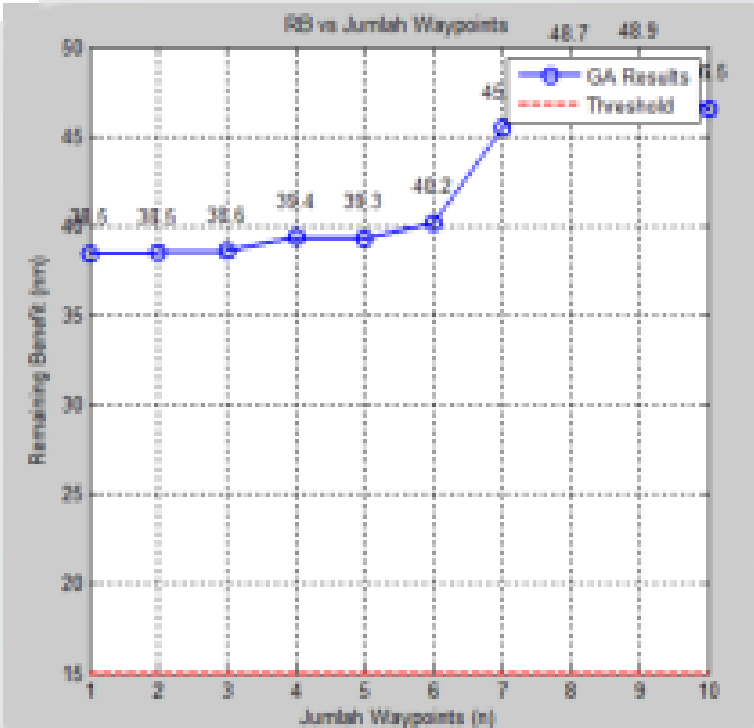


Tabel 4.1.1: Rekapitulasi Waypoint, Radius Belok, Sudut, dan Koordinat

Waypoint	RB (nm)	Generasi	Titik	Lintang (°)	Bujur (°)
1	38.46	1000	X1	-33.9754	25.6931
2	38.54	1000	X1	-1.1972	99.9146
			X2	-1.1909	99.9289
3	38.60	1000	X1	-1.3280	99.6185
			X2	-1.3103	99.6585
			X3	-1.2718	99.7456
4	39.38	1000	X1	-33.9012	25.8610
			X2	-2.4723	97.0274
			X3	-2.3984	97.1945
			X4	-1.1292	100.0686
5	39.32	1000	X1	-33.9701	25.7052
			X2	-2.5866	96.7684
			X3	-2.4085	97.1717
			X4	-1.3130	99.6524
			X5	-1.2488	99.7977
6	40.18	1000	X1	-33.9684	25.7089
			X2	-4.1012	93.3388

Waypoint	RB (nm)	Generasi	Titik	Lintang (°)	Bujur (°)
			X3	-3.2930	95.1691
			X4	-3.2602	95.2431
			X5	-3.2545	95.2561
			X6	-1.0475	100.2536
7	45.48	1000	X1	-33.9635	25.7200
			X2	-33.9548	25.7397
			X3	-13.1234	72.9094
			X4	-13.0433	73.0909
			X5	-13.0363	73.1066
			X6	-1.0704	100.2016
			X7	-1.0370	100.2774
8	48.70	1000	X1	-21.3590	54.2611
			X2	-21.2664	54.4708
			X3	-21.2530	54.5012
			X4	-19.3127	58.8946
			X5	-19.3046	58.9131
			X6	-1.5397	99.1391
			X7	-1.1347	100.0561
			X8	-1.0386	100.2738
9	48.91	1000	X1	-33.9714	25.7022
			X2	-33.9158	25.8281
			X3	-25.5036	44.8763

Waypoint	RB (nm)	Generasi	Titik	Lintang (°)	Bujur (°)
			X4	-24.9776	46.0673
			X5	-24.9529	46.1233
			X6	-7.1700	86.3901
			X7	-7.1168	86.5106
			X8	-6.4641	87.9885
			X9	-6.4572	88.0042
10	46.54	1000	X1	-15.0718	68.4976
			X2	-14.8292	69.0469
			X3	-12.6320	74.0223
			X4	-12.5747	74.1520
			X5	-12.5346	74.2426
			X6	-12.5204	74.2749
			X7	-12.3638	74.6294
			X8	-2.3785	97.2396
			X9	-2.3040	97.4085
			X10	-1.1046	100.1243



Gambar 4.1.1: RB akibat pemilihan *waypoint*

Key Achievements:

- Developed an optimization framework for maritime routing using Genetic Algorithms.
- Applied elliptic and spherical geometry concepts to real-world navigation problems.
- Performed numerical simulations for international maritime routes across the Indian Ocean.
- Identified that using 1–2 waypoints provides the most efficient routing performance with lower Remaining Benefit (RB) values.
- Evaluated the trade-off between waypoint quantity and route efficiency.

Numerical Process Using MATLAB

[Click here for detail](#)



Research Title: Mathematical Modeling and Stability Analysis of Rumor Spreading Dynamics Using the SPIUH Model

Brief Description:

Developed a mathematical model to analyze rumor spreading dynamics in social networks using a nonlinear system of differential equations. The study applied stability analysis, Jacobian matrices, eigenvalue analysis, and the basic reproduction number (R_0) to evaluate rumor propagation and control strategies.

Topics & Expertise:

Mathematical & Analytical Skills:

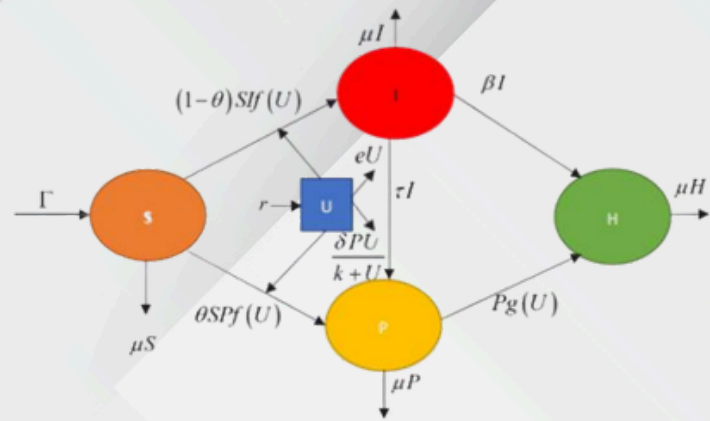
- Mathematical Modeling
- Differential Equations
- Dynamical Systems
- Stability Analysis
- Nonlinear System Analysis
- Eigenvalue & Eigenvector Analysis
- Jacobian Matrix Analysis
- Quantitative Research

Computational & Research Skills:

- Numerical Simulation
- Rumor Propagation Modeling
- Epidemic-inspired Modeling
- Equilibrium Analysis
- Next Generation Matrix (NGM)
- Routh–Hurwitz Stability Criterion

Methods & Techniques Used:

- Ordinary Differential Equation (ODE) Modeling
- Stability Analysis of Equilibrium Points
- Jacobian Matrix Construction
- Basic Reproduction Number (R_0) Analysis
- Next Generation Matrix (NGM) Method
- Routh–Hurwitz Stability Criterion
- Numerical Verification using MATLAB



Gambar 5.1.1 : Diagram Skematik Alur Penyebaran Rumor

$$\begin{aligned} \frac{dS}{dt} &= \Gamma - ((1 - \theta)I + \theta P)Sf(U) - \mu S \\ \frac{dP}{dt} &= \theta SPf(U) + \tau I - (\mu + g(U))P \\ \frac{dI}{dt} &= (1 - \theta)SI f(U) - (\mu + \tau + \beta)I \\ \frac{dU}{dt} &= r - eU - \frac{\delta PU}{K + U} \\ \frac{dH}{dt} &= \beta I + Pg(U) - \mu H \end{aligned}$$

• Titik Ekuilibrium Bebas Rumor didefinisikan sebagai:

$$E_0 = \left(\frac{\Gamma}{\mu}, 0, 0, \frac{r}{e}, 0\right)$$

$$J(E_0) = \begin{bmatrix} -\mu & -\theta \frac{\Gamma}{\mu} f\left(\frac{r}{e}\right) & -(1-\theta) \frac{\Gamma}{\mu} f\left(\frac{r}{e}\right) & 0 & 0 \\ 0 & \theta \frac{\Gamma}{\mu} f\left(\frac{r}{e}\right) - \left(g\left(\frac{r}{e}\right) + \mu\right) & \tau & 0 & 0 \\ 0 & 0 & (1-\theta) \frac{\Gamma}{\mu} f\left(\frac{r}{e}\right) - (\mu + \tau + \beta) & 0 & 0 \\ 0 & \frac{-\delta r}{e\left(k + \frac{r}{e}\right)} & 0 & -e & 0 \\ 0 & g\left(\frac{r}{e}\right) & \beta & 0 & -\mu \end{bmatrix}$$

Simulasi Numerik

Dari hasil data yang diperoleh didapatkan nilai dari setiap parameter yaitu sebagai berikut:

No	Parameter	Nilai	Sumber
1	Γ	0,25635	Data
2	δ atau θ	0,05323	Data
3	μ atau τ	0,3462	Data
4	β	0,2371	Data
5	r	0,94117	Data
6	e	0,704545	Data
7	K	1	Data
8	$f(U)$	0,69692	Data
9	$g(U)$	0,0964696	Data
10	S_0	40	Data
11	P_0	19	Data
12	I_0	14	Data
13	U_0	0,74047	Data
13	H_0	14	Data

Titik Ekuilibrium Bebas Rumor:

$$E_0 = \left(\frac{\Gamma}{\mu}, 0, 0, \frac{r}{e}, 0\right)$$

Titik Ekuilibrium Endemik Rumor:

$$\begin{aligned} S_1 &= -\frac{((1 - \theta)I_1 + \theta P_1)}{1 - \theta} + \frac{\Gamma}{\mu} \\ P_1 &= \frac{(eU_1 - r)(K + U_1)}{\delta U_1} \\ I_1 &= \frac{-\theta S_1 P_1 f(U_1) + (\mu + g(U_1))P_1}{\tau} \\ H_1 &= \frac{\beta I_1 + P_1 g(U_1)}{\mu} \end{aligned}$$

Key Achievements:

- Developed a nonlinear mathematical model for rumor propagation dynamics using the SPIUH compartment framework.
- Applied stability analysis, Jacobian matrices, and eigenvalue methods to evaluate system behavior and equilibrium stability.
- Calculated the basic reproduction number ((R_0)) using the Next Generation Matrix (NGM) approach to determine rumor transmission conditions.
- Constructed and analyzed systems of ordinary differential equations (ODEs) to model interactions within social network populations.
- Implemented the Routh–Hurwitz criterion and numerical simulations to validate the stability and reliability of the proposed model.
- Demonstrated strong quantitative research, mathematical modeling, and analytical problem-solving skills through computational and theoretical analysis.

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Let's work together



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Hazem Hawari



Padang, Sumatera Barat, Indonesia